CSE 370– Database Systems

Assignment 2

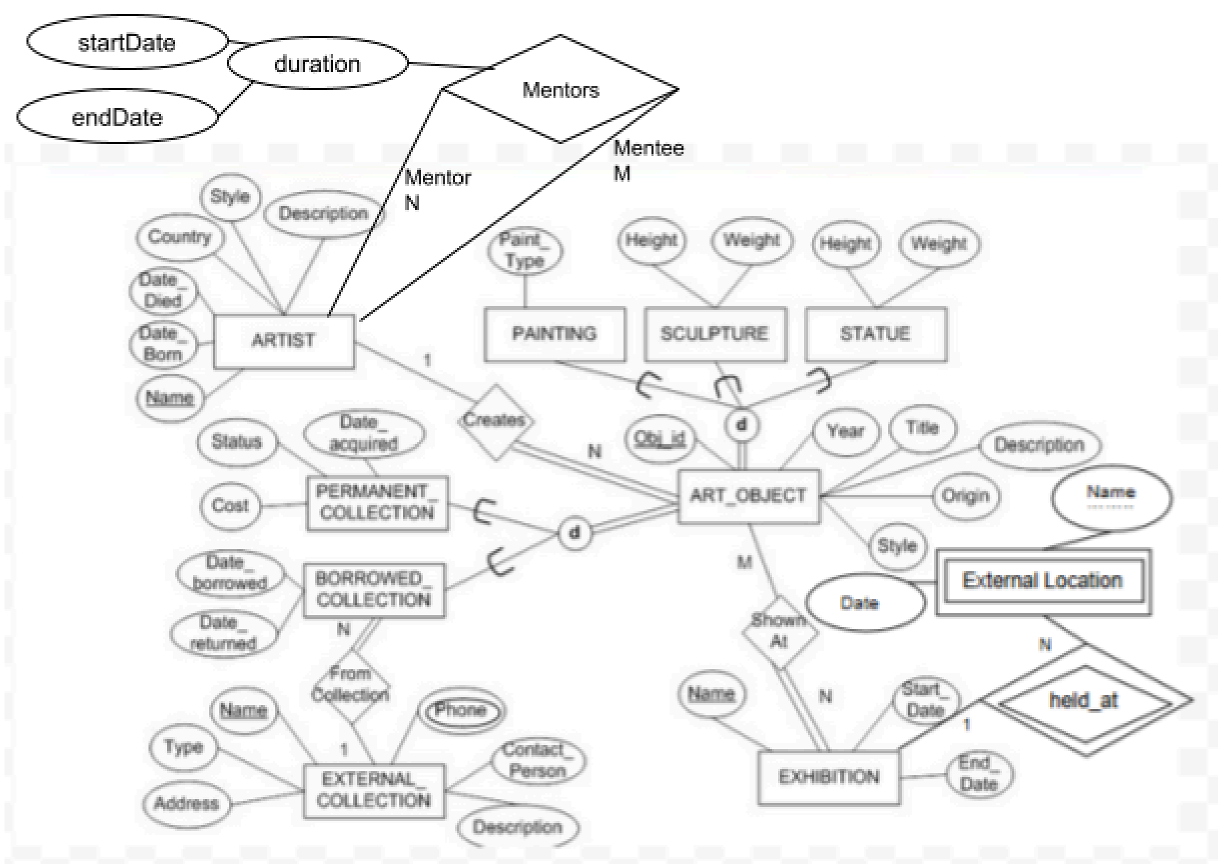
Spring 2025

Submission Instructions:

1. Write your name, id, section on top of the first page
2. Your answer should be handwritten, take pictures and create a single pdf.
3. **Submit the pdf in the following form**: https://forms.gle/s269cy4H2ohXfQy66
4. **Submission Deadline**: 26th April 2025 (Saturday), 11:59 pm (midnight)
5. **NO LATE SUBMISSION WILL BE ACCEPTED**

Question 1 **[12+ 2 = 14 Marks]**

a. You have to **apply** appropriate rules to map the following Extended Entity Relationship (EER) diagram to a relational model schema. Use two different options for mapping the two specialization/generalizations in the diagram. You may choose any two suitable options.



b. For the specialization/generalizations in the above diagram, which option between 8C (single table with 1 type attribute) and 8D(single table with multiple type attributes) is more efficient? Explain your answer.

Question 2 **[4 X 2 =8 Marks]**

X	Y	Z	A
abcefg	1	q	10
xyz	2	p	11
feg	3	q	12
xyz	2	p	13
abcdefg	3	q	10

State which of the following dependencies are valid and which are not. For each dependency, briefly write the reasons. The first one is explained and solved for you:

A. XY -> A

Functional dependency is valid if for the same value of XY, the corresponding value of A is also the same. Here (“xyz”, 2) is the repeated value for X and Y in row 2 and 4. The corresponding value in A is 11 and 13 respectively, therefore this functional dependency is not valid according to the definition of FD.

[Note: if all values of XY were unique, i.e. no repetition, then the FD will be valid as it does not contradict the definition of FD]

B. X -> YZ

C. A -> XYZ

D. YZ -> X

E. Y -> Z

Question 3 **[2+3+3 = 8 Marks]**

Consider the following relation:
Pokemon_Battles (Battle_id, Trainer_id, Battle_name, Trainer_Name, battle_date, {rewards}, Pokemon_id, pokemon_name, type, weakness, score)
Note: Multivalued is shown in curly braces{}. The primary key of the relation is underlined.
The relation has the following additional functional dependencies:
FD1: Battle_id□ Battle_name, battle_date
FD2: Trainer_id□ Trainer_Name

FD3: Pokemon_id→pokemon_Name, type, weakness
FD4: type→weakness

a. Explain if the above relation in first normal form (1NF) or not? If not, **apply** 1NF normalization.
b. Explain if the schema in (a) is in the second normal form (2NF) or not? If not, **apply** 2NF normalization.
c. Explain if the schema in (b) is in the third normal form (3NF) or not? If not, apply 3NF normalization.

Show full schema after each normalization step.